

- 9 (a) Students from three classes A, B and C took a Mathematics examination and their mean marks and standard deviation were recorded in the table.

| Class | Number of students | Mean mark | Standard deviation |
|-------|--------------------|-----------|--------------------|
| A     | 15                 | 85.4      | 0.313              |
| B     | 10                 | 86.5      | 2.128              |
| C     | 14                 | 75        | 0                  |

- (i) State how it may be deduced from the data that the mark scored by each student in class C was 75 marks.

Answer \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_ [2]

- (ii) Compare the performance of the classes.

Answer \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_ [2]

- (iii) A new student Raymond joined class C and the standard deviation of the marks of class C changed to 2.30. Find the possible values of Raymond's marks, giving your answer correct to the nearest whole number.

Answer \_\_\_\_\_ or \_\_\_\_\_ [4]

- (b) A box contains 8 cards, numbered '1' to '8' respectively. A card is drawn with replacement from the box until an '8' is obtained. Find the probability that

(i) the first draw was **not** an '8',

Answer \_\_\_\_\_ [1]

(ii) it will take exactly  $n$  draws to obtain an '8', giving your answer in terms of  $n$ ,

Answer \_\_\_\_\_ [1]

(iii) it will take at least  $n$  draws to obtain an '8', giving your answer in terms of  $n$ .

Answer \_\_\_\_\_ [1]

- (c) Krystal said that since she has not drawn an '8' for the last 15 draws, her next draw will likely be an '8'. Comment on the validity of her statement.

Answer \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_ [2]

**10** The volume of mixture required to make one cookie is  $15 \text{ cm}^3$ .

- (a)** The mixture is first rolled into a sphere before baking.  
Calculate the radius of the sphere.

Answer \_\_\_\_\_ cm [2]

- (b)** After it is baked, the cookie takes the shape of a cylinder of radius 3 cm and height of 6 mm due to air trapped in the cookie during baking.

- (i)** Calculate the volume of air trapped in the cookie.

Answer \_\_\_\_\_  $\text{cm}^3$  [2]

- (ii)** Express this volume of air as a percentage of the total volume of the cookie.

Answer \_\_\_\_\_ % [1]